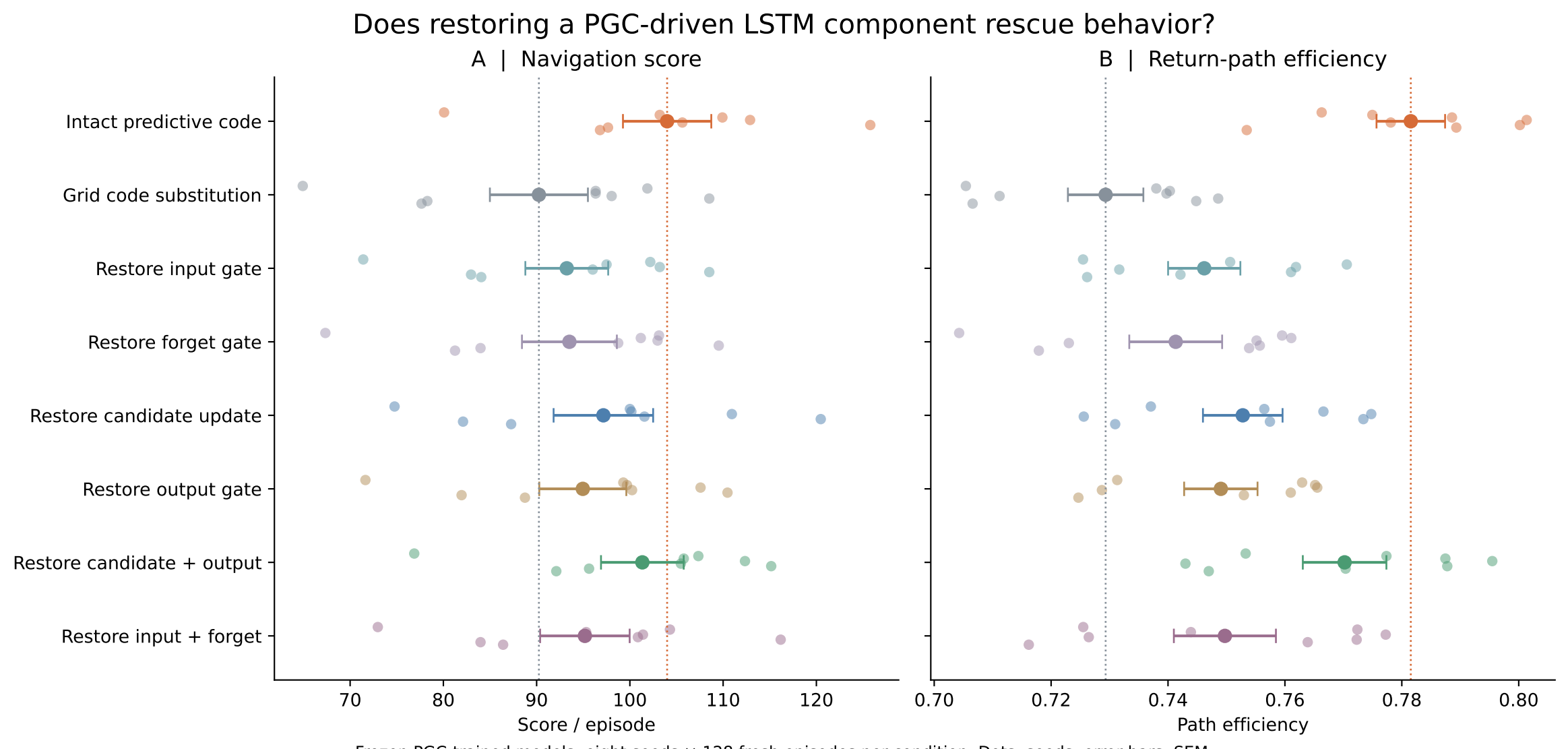
Does restoring a PGC-driven LSTM component rescue behavior?
A | Navigation score
B | Return-path efficiency
Intact predictive code
Grid code substitution
Restore input gate
Restore forget gate
Restore candidate update
Restore output gate
Restore candidate + output
Restore input + forget
70
80
90
100
110
120
Score / episode
0.70
0.72
0.74
0.76
0.78
0.80
Path efficiency
Frozen PGC-trained models; eight seeds × 128 fresh episodes per condition. Dots: seeds; error bars: SEM.
Components not restored receive grid-code drive; all conditions keep their own evolving recurrent state and behavior.